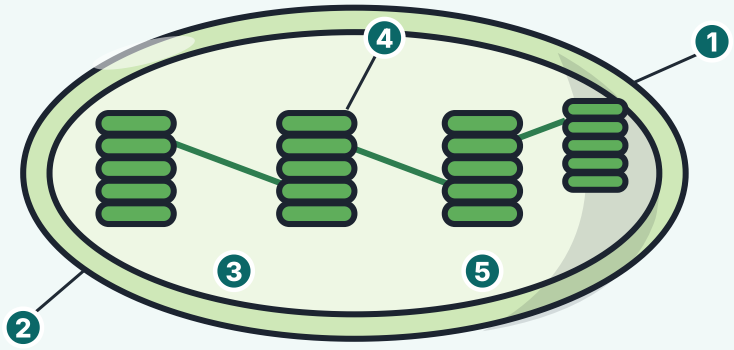
PHOTOSYNTHESIS

WHERE IT HAPPENS · THE CHLOROPLAST



1. Outer membrane
2. Inner membrane
3. Stroma: **Calvin cycle**
4. Thylakoid: **light reactions**
5. Granum (stack of thylakoids)

OVERALL EQUATION · LIGHT ENERGY

$$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$$

Simplified cross-section, not to scale.

THE TWO STAGES

CRITERION	LIGHT REACTIONS	CALVIN CYCLE
Where	thylakoid membrane	stroma
Needs	light, H_2O	CO_2, ATP, NADPH
Makes	ATP, NADPH, O_2	G3P (→ glucose)
Key point	O_2 comes from **water**	RuBisCO fixes CO_2

ENERGY FLOW

LIGHT absorbed by chlorophyll → **ATP + NADPH** chemical energy → **SUGAR** CO_2 fixed

KEY POINT

The light reactions turn light into **chemical energy** (ATP, NADPH); the Calvin cycle **spends** it to build sugar from CO_2.

MUST KNOW

MEMORIZE

- ☑ Chlorophyll absorbs mostly **blue** and **red**; reflects green.
- ☑ Water splitting (photolysis) releases O_2.
- ☑ Limiting factors: light, CO_2, temperature.



✘ COMMON TRAP

× "The Calvin cycle only happens in the dark." → **It does not need light directly, but depends on ATP/NADPH from the light reactions.**

× "Plants photosynthesize instead of respiring." → **Plants do both.**

TIP

Follow the atoms: the C and O in glucose come from **CO_2**; the O_2 released comes from **H_2O**.

References: OpenStax Biology 2e, 8.1 Overview of Photosynthesis; 8.2 The Light-Dependent Reaction; 8.3 Using Light Energy to Make Organic Molecules.